

02-719 Week 4

Genomics and Epigenetics of the Brain

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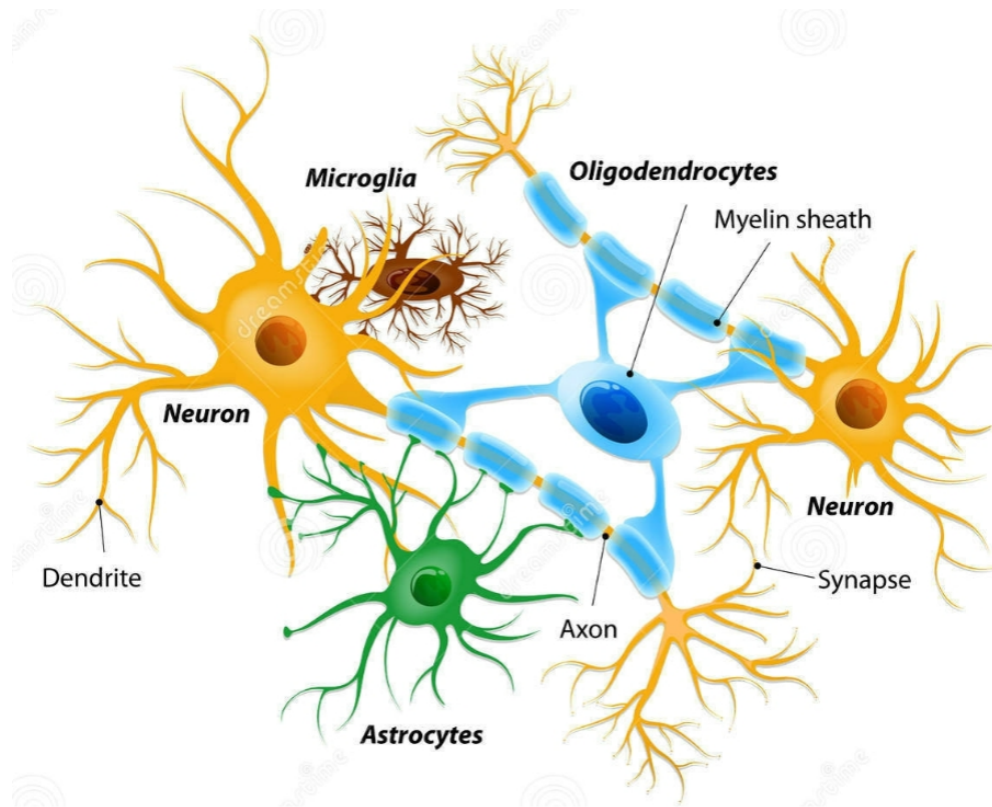
Diversity of Neuron Subtypes

Neuroscience in 2024:

- 15+ types of cortical interneurons with distinct morphologies, firing patterns, and connectivity
- 49 cortical cell populations with distinct transcriptomes
- Just beginning to understand the diversity of neurons on the epigenetic level and we are seeing vastly different epigenetic landscapes

Major Cell Types of the Brain

- Oligodendrocytes
 - Form the myelin sheath around axons, allowing for fast signal propagation
- Microglia
 - Immune cells that detect damaged or unhealthy neurons and protect against foreign bacteria and viruses
- Astrocytes
 - Provide structure to hold neurons in place
 - Supply nutrients
 - Digest debris from dead cells
- Neurons
 - Facilitate information transfer through propagation of electrical signal

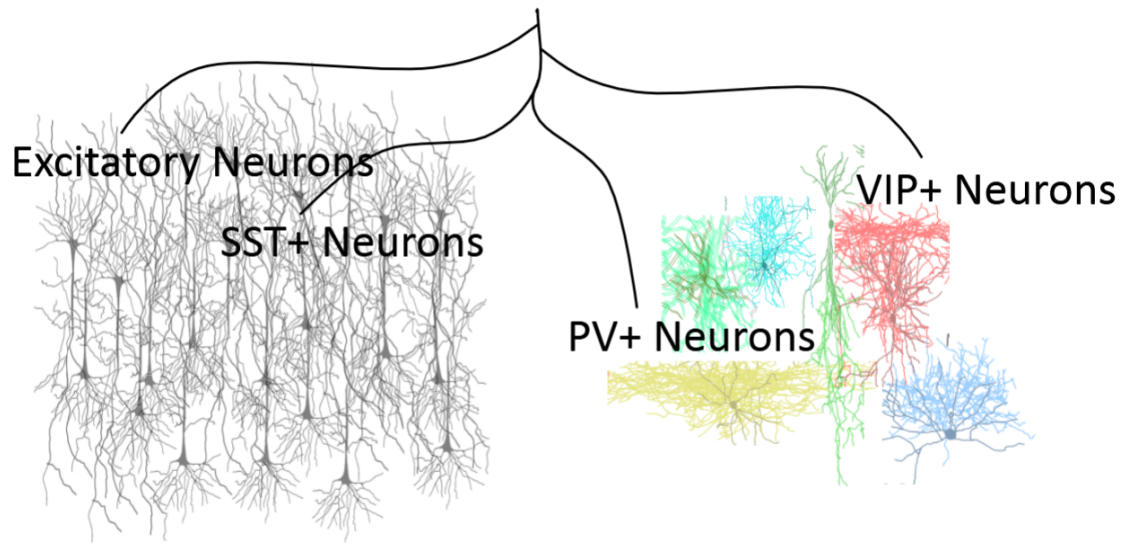


Types of Neurons

Neurons can be divided into two groups:

- **Glutamatergic Pyramidal Neurons:** excitatory neurons.
 - Use glutamate as a chemical messenger to increase brain activity. Receiving cells are more likely to fire an electrical signal
 - Drive most rapid signaling, thinking, learning, and memory.
 - Hyperactive glutamatergic neurons leads to conditions like epilepsy.
- **GABAergic Interneurons:** inhibitory neurons
 - Use gamma-aminobutyric acid (GABA) as a chemical messenger to calm brain activity. Receiving cells are less likely to fire an electrical signal
 - Prevent runaway seizures and balance brain signals.

Each group contains more different types of neurons.



Cellular Diversity in the Cortex

Excitatory Neurons: “Principal Cells”

- **Intracortical:**

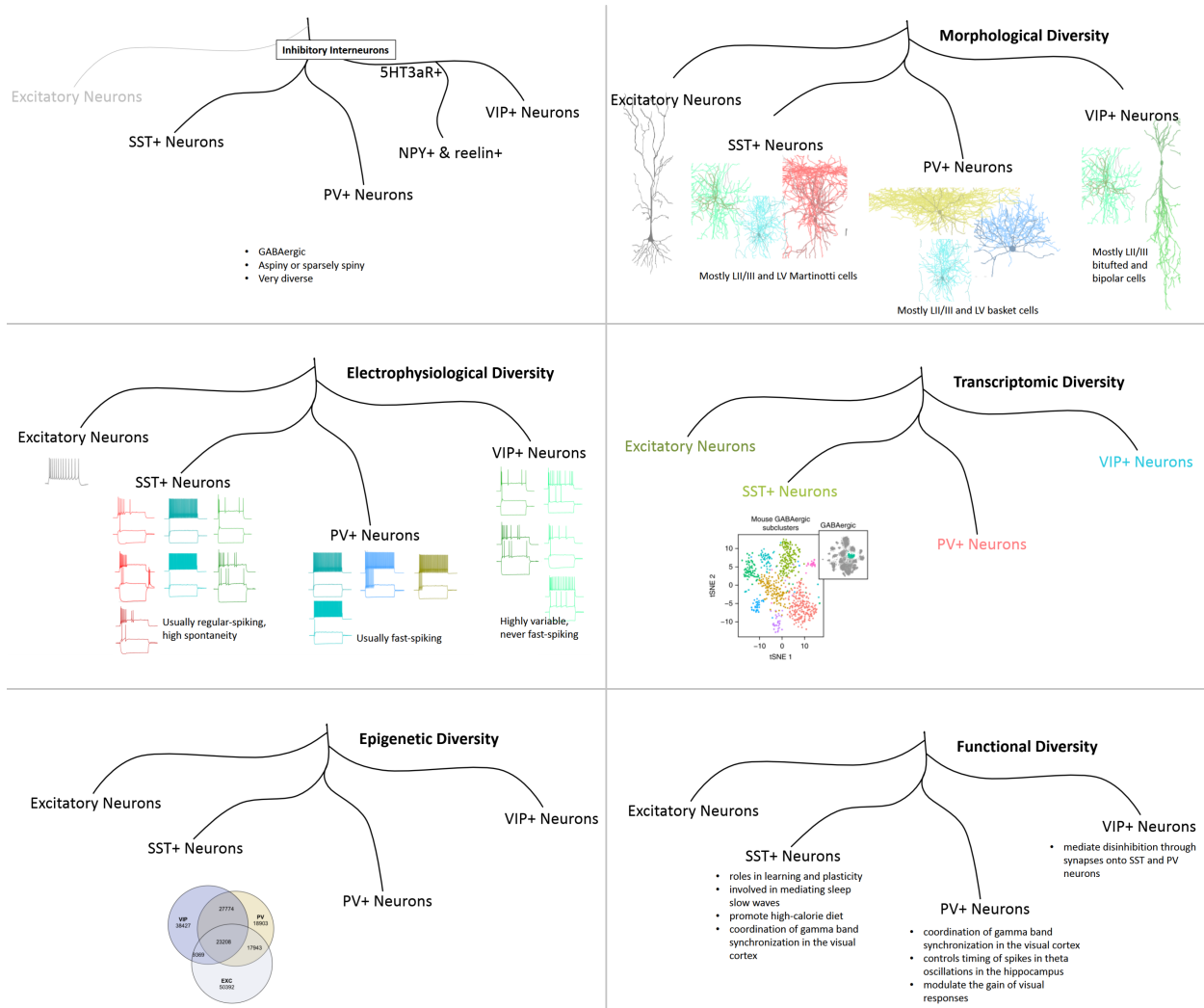
- Predominantly in Layer 2 / 3
- Projections stay within the cortex
- Can be *associative* (project to the same hemisphere, area, or column that the cell body is in) or *commissural* (project to the opposite hemisphere)

- **Corticofugal:**

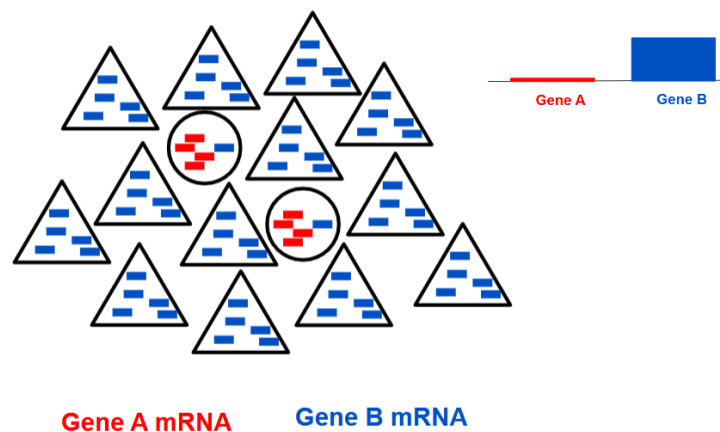
- Mainly in Layer 5 / 6
- Project to brain regions outside of the cortex
- Can be *corticalthalamic* (project to the thalamus, important for processing sensory information) or *subcerebral* (project to the brainstem or spinal cord)

Molecular interrogation reveals additional heterogeneity within subtypes of excitatory neurons. Subtypes can be defined by characteristic expression levels of a set of genes

Different Views of Diversity



Issues with Bulk Tissues



Signals from rare cell types can often be missed entirely.

Searching for the Genetic Basis of Disease

- We can run GWAS (genome-wide association study) to correlate SNPs with certain diseases.
- With enough reads, we can determine which peaks (SNPs) represent mutations that correlate with the presence of diseases
- Cell type-specific peaks overlap key disease SNPs, which might lead to problems. Is a mutation a cause for a disease, or is it normal for a different cell type?

Single Cell Experiments

To do sequencing studies, the vast majority of the data comes from single cell experiments.

Tissue Source

- **Fresh tissue:**
 - Best cell viable - best quality RNA
 - Requires direct access to tissue, lower throughput
- **Young (embryonic) tissue:**
 - Less myelinated tissue to dissociate
 - Age is experimental confounder
- **Cryo-preserved frozen:**
 - Better cell viability, higher throughput
 - Freeze-thaw effects in sample
- **Flash frozen**
 - Human brain repositories usually flash freeze tissue
 - Worst cell viability

Enzymatic Dissociation of Single Cells

1. Tissue is dissected and cut into thin slides.
2. Tissue is rinsed with PBS (Phosphate Buffered Saline), then minced.
3. Digestive enzymes are added (which might include collagenase, hyaluronidase, DNase, Papain, etc.)
4. Mixture is incubated to ensure tissue is digested
5. Sample is inspected and filtered (so only nuclei are left)
6. Nuclei can be input through a machine for cell counting and sequencing

Differences Between Single Cell and Single Nuclei

As it turns out, the amount of RNA from cells is more than the RNA in nuclei

- This is because RNA is present in cytoplasm

However, the number of cell types detected from cells is equal to types detected from nuclei

- Essentially, a cell's DNA sequence (and epigenetic markers) can be used to determine cell type

Both preparations can detect similar cell types.

Fresh Nuclei from Fresh Brain Tissue

- Less size bias in nuclei vs. cells
- Pestle mechanically break up extracellular matrix
- Iso/hypotonic buffer solution maintain organelle membranes (e.g., nuclei, mitochondria, etc.)

Nuclei are then purified using a density gradient.

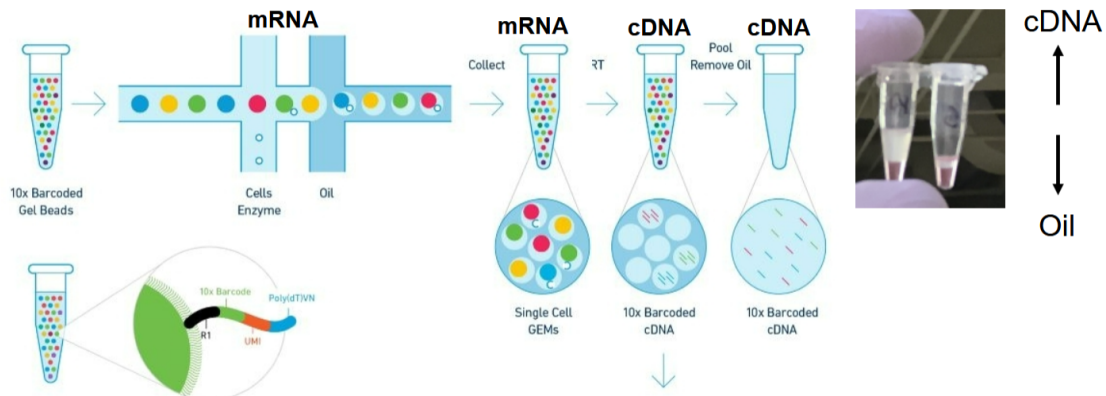
- Use solutions of 40% iodixanol, 30% iodixanol, and 25% iodixanol. Higher percentages are denser than lower percentages, and will therefore sink.
- Nuclei, being more dense, will sink to between the 30% and 40% layers. Meanwhile, all the junk will float at the 25% layer.
- A needle can be used to extract the nuclei from the middle layers.

Afterwards, the nuclei can be stained and counted using a machine.

Cell Counting with Microfluidics

1. Enzymes and nuclei are first loaded into microfluidics chip
2. Nuclei and barcode beads are partitioned with 10x controller (run through machine)
3. Droplets are suspended in oil.
4. DNA in droplets are amplified then sequenced.

What is Happening in the Microfluidics Black Box

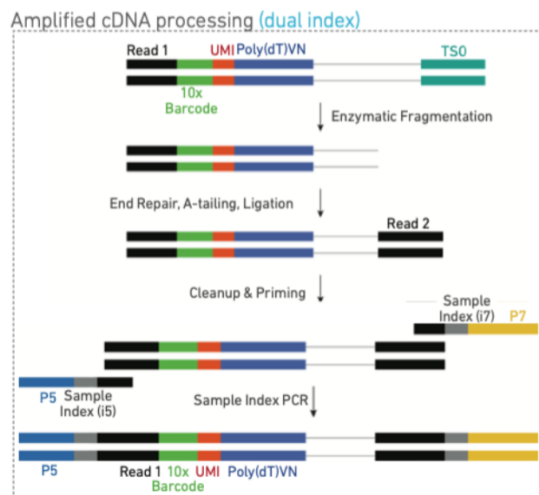
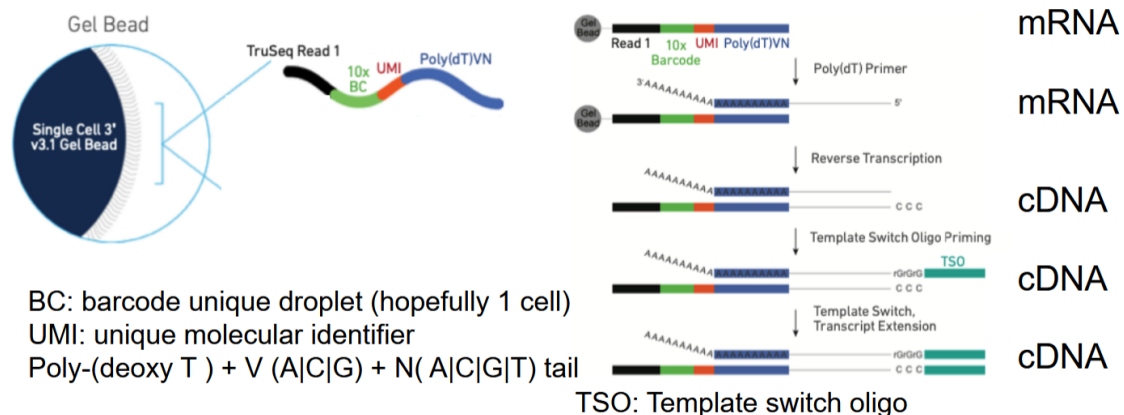


- Essentially, each gel bead comes with a barcode attached to it. A barcode is usually around 10 nucleotides long.
- Each gel bead is placed with a cell and enzymes, in a droplet. (Droplet is water floating in oil.)
- The enzyme will slowly integrate the barcode DNA into the DNA of the cell.
- Therefore, when sequencing, it becomes possible to count DNA.

Issues

- Depending on the quality of the sample, the cell in the droplet might be lysed. As a result, there could be stray RNA floating around in the droplet but not in the nuclei. The oil needs to be separated out before sequencing for this reason.
- Some gel beads might have two nuclei (by accident). Due to the nature of the process and how small individual droplets are, there isn't a way to detect this. When analyzing the data it needs to be taken into account.
- Barcodes attached to two nuclei means that some "cells" that are detected might have more genes than normal.

Tracking the RNA Molecule



BC: barcode unique droplet (hopefully 1 cell)

UMI: unique molecular identifier

Poly-(deoxy T) + V (A|C|G) + N (A|C|G|T) tail

TSO: template switch oligo

Read 2: cDNA sequence

Sample Index (i5/7): barcode tissue

P5/7 index: compatible with Illumina sequencing

Conclusions

- sc/snRNA-seq require many experimental considerations
 - Tissue, dissociation, etc.
- Different methods of labeling cells, RNA molecules
- Set up of barcode/UMI/cDNA fragment determine downstream post-sequencing analysis pipeline